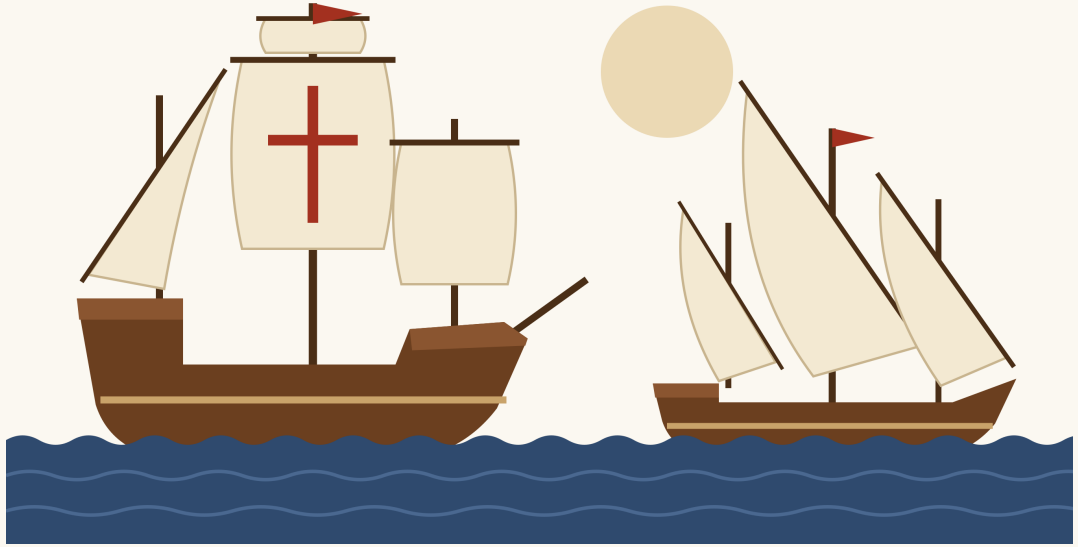
DEAPSECURE · BIG DATA · WARM-UP

Your First DataFrame

with the four voyages of Christopher Columbus, 1492–1504

A short pandas exercise before Session 1, using small tables you can check by eye.



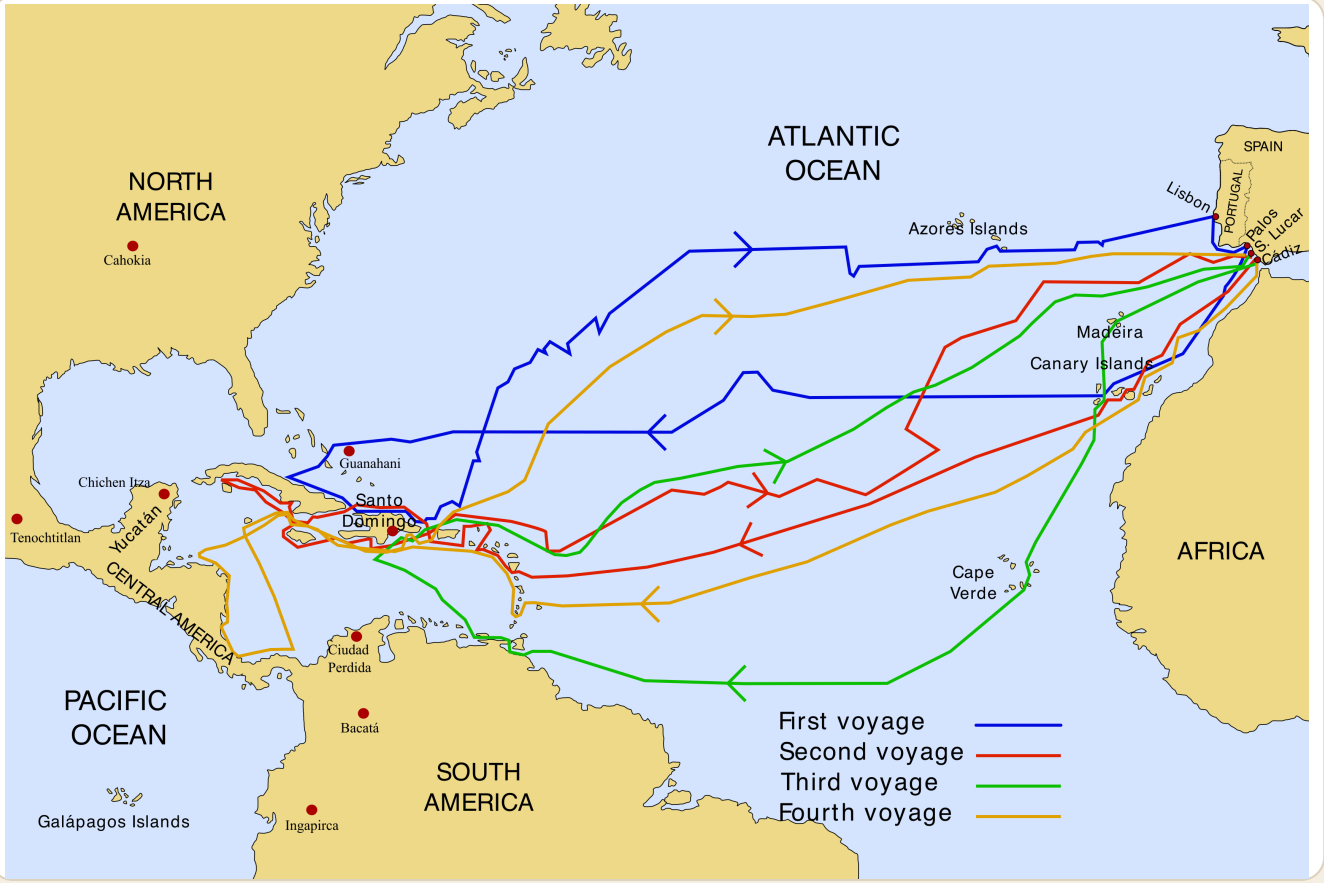
Carrack: large, high castles (the Santa María)

Caravel: small and fast (the Pinta, the Niña)

Notebook: BigData-warmup-columbus.ipynb · Slides: slides/Pandas-Warmup-Columbus.pdf (also .pptx)

THE VOYAGES

Four voyages across the Atlantic, 1492–1504



4 voyages

30 ships in total (3 + 17 + 6 + 4)

12 years, from 1492 to 1504

21 main landfalls in our table

Map: Phirosiberia, Wikimedia Commons, CC BY-SA 3.0

WHAT YOU WILL PRACTICE

Eight pandas skills on one small dataset

1

Create a DataFrame

`pandas.DataFrame()`

2

Inspect it

`df.shape, df.dtypes`

3

Select columns and rows

`df["col"], df.loc[2]`

4

Filter rows

`df[df["col"] == x]`

5

Find missing values

`df.isna().sum()`

6

Sort and add columns

`sort_values()`

7

Group and count

`groupby().count()`

8

Plot and map

`barh(), line_geo()`

A dictionary becomes a table

```
VOYAGES = pandas.DataFrame({
  "voyage":      [1, 2, 3, 4],
  "depart_port": ["Palos", "Cadiz", "Sanlucar", "Cadiz"],
  "num_ships":   [3, 17, 6, 4],
  # ... depart_date, return_date, first_landfall
})
```

	voyage	depart_port	num_ships
0	1	Palos	3
1	2	Cadiz	17
2	3	Sanlucar	6
3	4	Cadiz	4

- Each **key** becomes a column label.
- Each **list** holds one value per row.
- pandas adds the **index** 0, 1, 2, 3 (in orange).

Five things to check on any new table

Code	What it tells you	For VOYAGES
`df.head()`	The first rows (5 by default)	All 4 voyages
`df.shape`	(rows, columns)	(4, 6)
`df.columns`	The column labels	voyage, depart_date, …
`df.dtypes`	The data type of each column	Dates are stored as text
`df.info()`	All of the above, plus missing values	No missing values

! **Watch the dates:** stored as text, they can't be subtracted to get a voyage's length. We fix this in Section 5.

Picking columns and rows

Code	Selects	Returns
`df["depart_port"]`	One column	A Series
`df[["voyage", "num_ships"]]`	Several columns	A DataFrame
`df.loc[2]`	The row with label 2	A Series (one row)
`df.iloc[0]`	The first row, by position	A Series (one row)

i **Double brackets** select several columns: the inner [ ] is a Python list of column names.

Which ships have a known captain?

```
# keep the rows where the test is True
SHIPS[SHIPS["ship_type"] == "caravel"]

# count the missing values per column
SHIPS.isna().sum()

# ships whose captain is unknown
SHIPS[SHIPS["captain"].isna()]
```

The article names the 4th voyage's ships but not their captains, so those cells are **NaN** (missing).

voyage	ship	captain
1	Santa Maria	Christopher Columbus
1	Pinta	Martin Alonso Pinzon
1	Nina	Vicente Yanez Pinzon
4	Capitana	Christopher Columbus
4	Gallega	NaN
4	Vizcaina	NaN
4	Santiago de Palos	NaN

From text to dates, and from dates to days

```
VOYAGES["depart_date"] = pandas.to_datetime(VOYAGES["depart_date"])
VOYAGES["return_date"] = pandas.to_datetime(VOYAGES["return_date"])
VOYAGES["duration_days"] = (VOYAGES["return_date"] - VOYAGES["depart_date"]).dt.days
VOYAGES.sort_values("duration_days", ascending=False)
```

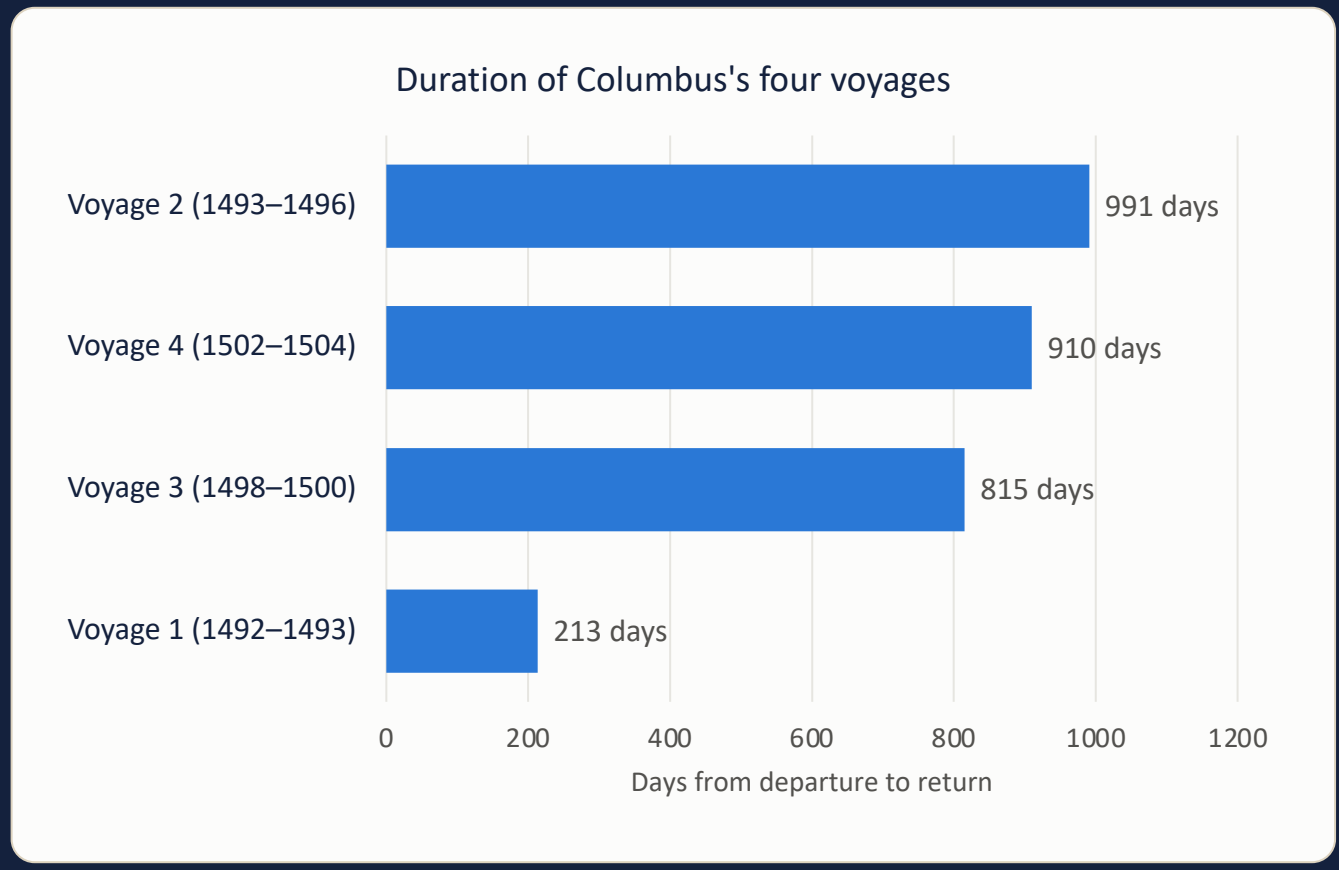
voyage	depart_date	return_date	duration_days
2	1493-09-24	1496-06-11	991
4	1502-05-12	1504-11-07	910
3	1498-05-30	1500-08-23	815
1	1492-08-03	1493-03-04	213

The 2nd voyage was the longest

991 days

The 1st voyage, the most famous, was the shortest: 213 days.

```
ax.barh(labels, plot_data["duration_days"])
```



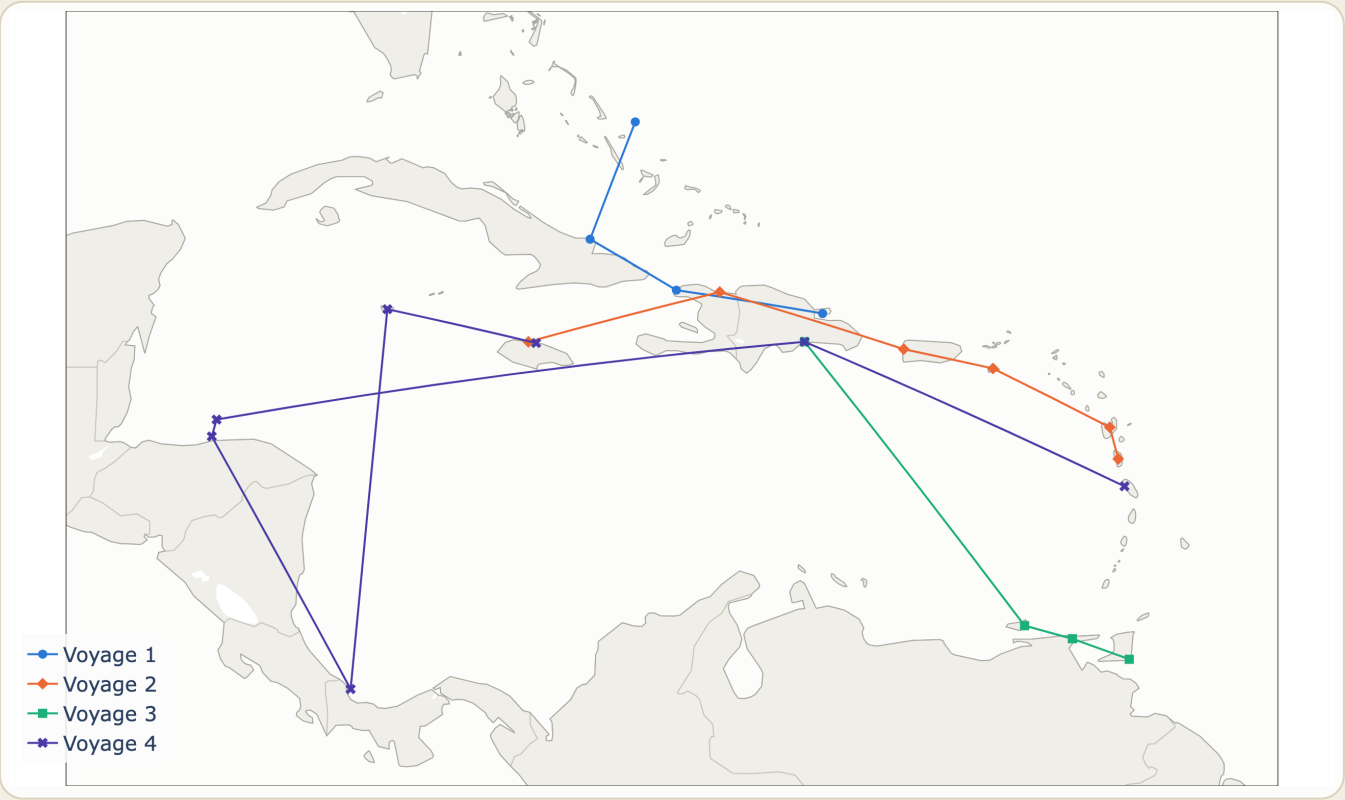
How many landfalls on each voyage?

```
LANDFALLS.groupby("voyage")["place"].count()
counts = LANDFALLS["place"].value_counts()
counts[counts > 1]    # places visited on more than one voyage
```

4	**6**	**4**	**7**
landfalls, voyage 1	landfalls, voyage 2	landfalls, voyage 3	landfalls, voyage 4

Hispaniola is the only place on more than one voyage: it appears 3 times.

Each row becomes a point on the map




```
px.line_geo(map_data,
  lat="lat", lon="lon",
  color="voyage",
  symbol="voyage",
  markers=True)
```

- **lat, lon:** where each point goes
- **color, symbol:** one style per voyage
- Lines join landfalls in date order, not the real routes

8 · BONUS: COMBINING TABLES

Joining each ship to its voyage

```
pandas.merge(SHIPS, VOYAGES[["voyage", "depart_port", "depart_date"]], on="voyage")
```

ship	voyage	depart_port	depart_date
Santa Maria	1	Palos	1492-08-03
Pinta	1	Palos	1492-08-03
Nina	1	Palos	1492-08-03
Capitana	4	Cadiz	1502-05-12
Gallega	4	Cadiz	1502-05-12
Vizcaina	4	Cadiz	1502-05-12
Santiago de Palos	4	Cadiz	1502-05-12

Voyages 2 and 3 are missing. Why?

Hint: look up the `how=` argument of pandas.merge.

YOUR TURN

Seventeen exercises, marked #TODO in the notebook

Section	Exercises
2 · Inspecting	2.1 shape · 2.2 data types
3 · Selecting	3.1 two columns · 3.2 one row with .loc
4 · Filtering	4.1 the caravels · 4.2 Columbus's ships · 4.3 count missing values · 4.4 unknown captains
5 · Dates and plots	5.1 duration_days · 5.2 sort by duration · 5.3 bar chart of num_ships
6 · Grouping	6.1 landfalls per voyage · 6.2 top countries · 6.3 repeated places · 6.4 challenge: first and last dates
7 · Mapping	7.1 map only the first voyage
9 · Excel and CSV	9.1 load landfalls.csv with read_csv and check an Excel answer

Suggested solutions and the Excel answers are at the end of the notebook. Try each exercise first.

TRY IT IN EXCEL

The same questions, answered in a spreadsheet

#	Question	File	In Excel
1	Which ships were caravels?	ships.csv	Filter
2	Which ships have no known captain?	ships.csv	Filter: (Blanks)
3	How many days did each voyage last? Sort longest first.	voyages.csv	Formula, then Sort
4	How many landfalls on each voyage?	landfalls.csv	COUNTIF or PivotTable
5	Which modern country appears most often?	landfalls.csv	PivotTable
6	Which place appears on more than one voyage?	landfalls.csv	COUNTIF
7	Make a bar chart of the voyage durations.	voyages.csv	Insert › Bar chart

! **Excel can't store dates before 1900,** so the dates open as text. For question 3, add 800 years (the leap years repeat every 400):

```
=DATE(LEFT(C2,4)+800,MID(C2,6,2),RIGHT(C2,2)) - DATE(LEFT(B2,4)+800,MID(B2,6,2),RIGHT(B2,2))
```

TRY IT IN EXCEL · ANSWERS

Answers to the Excel questions

#	Question	Answer
1	Which ships were caravels?	Pinta, Nina, Capitana, Gallega, Vizcaina, Santiago de Palos (6 of 7)
2	Which ships have no known captain?	Gallega, Vizcaina, Santiago de Palos
3	How many days did each voyage last?	Voyage 2: 991 · voyage 4: 910 · voyage 3: 815 · voyage 1: 213
4	How many landfalls on each voyage?	Voyage 1: 4 · voyage 2: 6 · voyage 3: 4 · voyage 4: 7
5	Which modern country appears most often?	Dominican Republic, 3 times
6	Which place appears on more than one voyage?	Hispaniola, 3 times
7	Bar chart of the voyage durations	The same ranking as the notebook's chart in Section 5

RECAP

A few commands cover most everyday work

`pandas.DataFrame()` `head()` `shape` `dtypes` `df["col"]` `.loc[ ]`

`df[condition]` `isna().sum()` `to_datetime()` `sort_values()`

`value_counts()` `groupby()` `merge()`

Next: Session 1. The same tools, on real system data.